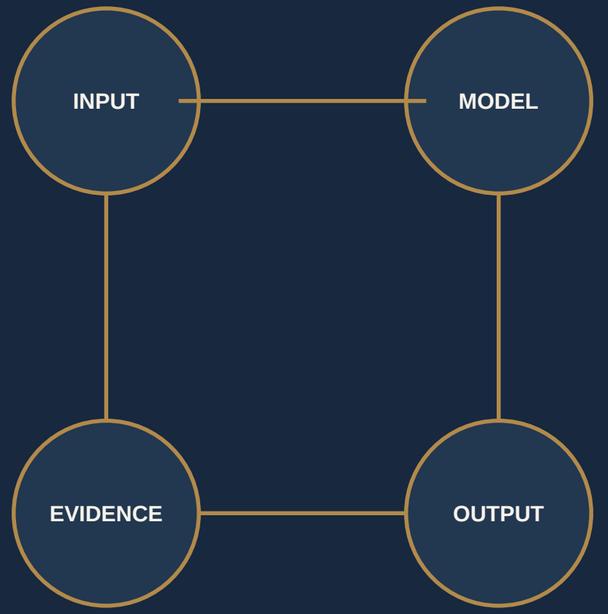
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Autonomous Systems

AI.204 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define autonomy architecture.
- Apply key methods including perception.
- Build or analyze a practical system involving localization.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Autonomy architecture

02

METHODS

Perception

03

SYSTEMS

Localization

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Autonomy architecture

Autonomy architecture

02 · Perception

Perception

03 · Localization

Localization

04 · Planning

Planning

05 · Control

Control

06 · Decision-making

Decision-making

07 · Uncertainty and safety

Uncertainty and safety

08 · Mandatory practical: Develop a

Mandatory practical: Develop a simulated autonomous robot or vehicle capable of sensing

Core ideas, made explicit

AI.204 • Distinguish definitions, assumptions and evidence

- Autonomy architecture
- Perception
- Localization
- Planning
- Control

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.204 • Distinguish definitions, assumptions and evidence

- Decision-making
- Uncertainty and safety
- Mandatory practical: Develop a simulated autonomous robot or vehicle capable of sensing
- Planning and acting in an environment

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Autonomy architecture

INPUTS
Perception

METHOD
Localization

OUTPUT
Planning

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving autonomy architecture.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using autonomy architecture.
- Implement or simulate a baseline using perception.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.